2 Harnessing the power of large language models

This chapter covers

- Understanding the basics of LLMs
- Connecting to and consuming the OpenAI API
- Exploring and using open source LLMs with LM Studio
- Prompting LLMs with prompt engineering
- Choosing the optimal LLM for your specific needs

The term *large language models* (LLMs) has now become a ubiquitous descriptor of a form of AI. These LLMs have been developed using generative pretrained transformers (GPTs). While other architectures also power LLMs, the GPT form is currently the most successful.

LLMs and GPTs are *generative* models, which means they are trained to *generate* rather than predict or classify content. To illustrate this further, consider figure 2.1, which shows the difference between generative and predictive/classification models. Generative models create something from the input, whereas predictive and classifying models classify it.